



Percutaneous coronary intervention versus coronary artery bypass grafting for unprotected left main stenosis: 10-year final results from the randomised, open-label, non-inferiority NOBLE trial

Emil Nielsen Holck, Niels R Holm, David Hildick-Smith, M Mitchell Lindsay, Mark S Spence, Andrejs Erglis, Ian B A Menown, Terje Steigen, Lone Juul Hune Mogensen, Ivy Susanne Modrau, Matti Niemelä, Jens F Lassen, Keith Oldroyd, Peteris Stradins, Simon J Walsh, Petter C Endresen, Thor Tørvik, Ole Frøbert, Alastair N J Graham, Vesa Anttila, Uday Trivedi, Leif Thuesen, Evald H Christiansen

Summary

Background Coronary artery bypass grafting (CABG) is recommended over percutaneous coronary intervention (PCI) for patients with significant unprotected left main coronary artery disease. We aim to provide long-term outcome data comparing PCI with newer generation drug-eluting stents and CABG, which are scarce.

Methods This previously published, prospective, randomised, open-label, non-inferiority trial enrolled patients with unprotected left main coronary artery stenosis at 36 hospitals in Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Norway, Sweden, and the UK. Eligibility was determined by a multidisciplinary heart team and defined by clinical criteria (chronic or acute coronary syndrome and a life expectancy of >1 year) and angiographic criteria (left main coronary artery diameter stenosis $\geq 50\%$ or fractional flow reserve ≤ 0.80 in the left main ostium, mid-shaft, or bifurcation). Patients with ST elevation myocardial infarction within 24 h or considered at too high risk for CABG or PCI were excluded. Patients with angiographically confirmed significant left main coronary artery disease were randomly assigned (1:1) to PCI or CABG using an online system and stratified by site, sex, distal left main coronary artery bifurcation lesions, and diabetes. The primary outcome was the difference in 10-year all-cause mortality in the intention-to-treat (ITT) population, which was analysed using Kaplan–Meier estimates and unadjusted Cox regression. Patients were censored at the date of death, emigration, withdrawal, or loss to follow-up. Variation in all-cause mortality was assessed in prespecified subgroups. The trial is registered with ClinicalTrials.gov, NCT01496651 (active, not recruiting).

Findings From Dec 9, 2008, to Jan 21, 2015, 1201 patients were randomly assigned to PCI (n=598) or CABG (n=603). 17 patients were lost to follow-up before 1 year. 592 patients in each group were included in the ITT population. Mean age was 66.2 years (SD 9.9) in the PCI group and 66.2 years (9.4) in the CABG group. 256 (22%) of 1184 participants were female and 928 (78%) were male. There was no difference in all-cause mortality at 10 years (136 [23%] of 592 in the PCI group and 145 [25%] of 592 in the CABG group; hazard ratio 0.93 [95% CI 0.74–1.18]; p=0.56). No significant difference in all-cause mortality with SYNTAX score was identified.

Interpretation There was no significant difference in all-cause mortality at 10 years between PCI and CABG for patients with unprotected left main coronary artery disease and no additional complex lesions, indicating that PCI is equally as safe as CABG in patients eligible for both treatments. These results will aid heart teams in developing an individualised patient-centred strategy.

Funding Biosensors and Aarhus University Hospital.

Copyright © 2026 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

Unprotected left main coronary artery disease is associated with high morbidity and mortality. Currently, coronary artery bypass grafting (CABG) is the recommended revascularisation strategy over percutaneous coronary intervention (PCI).^{1,2} Revascularisation with PCI has shown inferiority to CABG at 5 years after treatment due to a higher rate of spontaneous myocardial infarctions and repeat revascularisations.^{3–8} In a

meta-analysis of randomised controlled trials comparing PCI with CABG, left main coronary artery PCI appears safe after 5 years without any significantly increased risk of all-cause mortality.⁶ This meta-analysis warrants caution because the 1–5-year landmark estimate was odds ratio (OR) 1.24 (0.99–1.56; p=0.063) with CABG as a reference, which was probably driven by the mortality signal with CABG observed in the EXCEL trial.⁶ These findings prompted the need for long-term

Lancet 2026; 407: 1374–82

See [Comment](#) page 1308

Department of Cardiology

(E N Holck MD PhD,

N R Holm MD,

L J Hune Mogensen MStat,

Prof E H Christiansen PhD) and

Department of Cardio-Thoracic

Surgery (I S Modrau Dr Med),

Aarhus University Hospital,

Aarhus, Denmark; Department

of Clinical Medicine, Aarhus

University, Aarhus, Denmark

(E N Holck, L J Hune Mogensen,

I S Modrau, Prof O Frøbert PhD,

Prof E H Christiansen); Sussex

Cardiac Centre, University

Hospitals Sussex, Brighton, UK

(Prof D Hildick-Smith MD);

Heart, Lung, and Diagnostic

Division, Golden Jubilee

University National Hospital,

Clydebank, UK

(M M Lindsay MD,

Prof K Oldroyd MD);

Department of Cardiology,

Royal Victoria Hospital, Belfast,

UK (Prof M S Spence MD,

S J Walsh MD); Latvian Centre of

Cardiology (Prof A Erglis PhD)

and Cardiac Surgery Centre

(Prof P Stradins PhD), Pauls

Stradins Clinical University

Hospital, Riga, Latvia;

Craigavon Cardiac Centre,

Southern Health and Social

Care Trust, Portadown, UK

(Prof I B A Menown MD);

Department of Cardiology,

University Hospital of

Northern Norway, Tromsø,

Norway (T Steigen PhD);

Medical Research Center,

University of Oulu, Oulu

University Hospital, Oulu,

Finland (M Niemelä PhD);

Department of Cardiology,

Odense University Hospital,

Odense, Denmark

(Prof J F Lassen PhD);

Department of Thoracic and

Vascular Surgery, University

Hospital of Northern Norway,

Tromsø, Norway

Research in context

Evidence before this study

We searched PubMed from Sept 1, 2021, to Oct 1, 2025, using the search terms “left main”, “percutaneous coronary intervention” OR “stent”, and “coronary artery bypass grafting” to identify randomised controlled trials comparing percutaneous coronary intervention (PCI) using drug-eluting stents (DES) with coronary artery bypass grafting (CABG) in the left main coronary artery. Studies should report at least 5 years of follow-up because long-term follow-up is mandated when comparing CABG and PCI. We found a systematic review and meta-analysis of the NOBLE, SYNTAX Extended Survival, EXCEL, and PRECOMBAT trials. In a meta-analysis of 4394 patients, spontaneous myocardial infarction and repeat revascularisation was higher in patients who received PCI than those who received CABG 5 years after random assignment, whereas procedural myocardial infarction and stroke was higher in those who received CABG. No difference was observed in all-cause mortality, but the 1–5-year landmark analysis was non-significantly in favour of CABG (odds ratio 1.24 [95% CI 0.99–1.56]; $p=0.063$) driven by the EXCEL trial. A subgroup analysis of 1305 patients in the PRECOMBAT and SYNTAX Extended Survival trial with 10-year follow-up did not reveal an increased risk of all-cause mortality in patients who received first generation DES compared with CABG.

Added value of this study

In this randomised trial, second generation DES did not show a higher mortality than CABG at 10 years in patients with

unprotected left main coronary artery disease who were eligible for both treatments. Landmark analysis between 0–5 years and 5–10 years did not show increased mortality at any timepoint. In a subgroup analysis, a possible differential treatment effect was observed in patients with acute coronary syndrome as PCI was associated with a lower risk of all-cause mortality than CABG. No heterogeneous treatment effect with SYNTAX score was found.

Implications of all the available evidence

The 10-year follow-up in this trial supports the extended findings from the SYNTAX Extended Survival and PRECOMBAT trials using first generation DES. Based on all-cause mortality, PCI with second generation DES is as safe as CABG. Consequently, the higher rate of spontaneous myocardial infarction and repeat revascularisations with PCI does not translate into higher mortality, which supports CABG as the preferred revascularisation option to improve major adverse cardiac events, but equipoise is evident for all-cause mortality. Caution is warranted for the treatment of patients with unprotected left main coronary artery disease and acute coronary syndrome as they might benefit more from PCI than CABG. Since both PCI and CABG are feasible, current evidence supports nuanced shared decision making with clear communication of risks and benefits.

(P C Endresen PhD); Department of Cardiology, Nordland Hospital Trust, Bodø, Norway (T Trovik PhD); Department of Cardiology, Örebro University Hospital, Örebro, Sweden (Prof O Fröbert); Department of Cardiac Surgery, Royal Victoria Hospital, Belfast, UK (A N J Graham MD); Heart Center, Turku University Hospital, University of Turku, Turku, Finland (V Anttila PhD); Royal Sussex County Hospital, Brighton, UK (U Trivedi MD); Department of Cardiology, Aalborg University Hospital, Aalborg, Denmark (L Thuesen DMSc)

Correspondence to: Prof Evald H Christiansen, Department of Cardiology, Aarhus University Hospital, 8200 Aarhus, Denmark evald.christiansen@rm.dk

safety results, which were partly addressed by a subgroup analysis of the PRECOMBAT (with follow-up in 96% of patients) and SYNTAX Extended Survival trials (with follow-up in 93%) that showed no mortality differences at 10 years in patients who received first generation drug-eluting stents (DES) compared with those who received CABG.^{9,10} In EXCEL and NOBLE, patients primarily received second generation DES and the results on 5-year all-cause mortality were conflicting.^{3,5}

We aimed to provide final 10-year follow-up on all-cause mortality in this NOBLE study by investigating PCI versus CABG in patients with unprotected left main coronary artery disease and no additional complex lesions.

Methods

Study design

This prospective, randomised, open-label, non-inferiority trial has been described in detail previously.^{3,4} Patients with unprotected left main coronary artery stenosis were enrolled at 36 hospitals in Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Norway, Sweden, and the UK. This trial was approved by the national or local ethics committees and the Danish Data Protection Agency and conducted in accordance with the Declaration of Helsinki.

All patients provided written informed consent before randomisation. The trial protocol is available in the appendix (pp 19–45). The trial is registered with ClinicalTrials.gov, NCT01496651 (active, not recruiting).

Participants

Eligibility was defined by clinical criteria (chronic or acute coronary syndrome and a life expectancy of >1 year) and angiographic criteria (left main coronary artery diameter stenosis $\geq 50\%$ or fractional flow reserve ≤ 0.80 in the left main ostium, mid-shaft, or bifurcation). Patients with ST elevation myocardial infarction within 24 h or considered at too high risk for CABG or PCI were excluded. Additional complex lesions or at least three non-complex lesions were considered as an angiographic exclusion criterion. The patients were reviewed by an in-hospital heart team comprising at least one interventional cardiologist and one cardiac surgeon, who should reach consensus that the patient was eligible for both PCI and CABG.

Randomisation and masking

Patients were randomly assigned (1:1) to PCI or CABG using an online system (TrialPartner; hosted and maintained by the Department of Clinical Medicine,

See Online for appendix

Aarhus University, Aarhus, Denmark). Randomisation was stratified by site, sex, distal left main coronary artery bifurcation lesions, and diabetes. Masking of patients or treating physicians was not ethically feasible.

Procedures

Complete revascularisation with PCI or CABG was the treatment goal in both groups. The study procedures were described in detail previously.⁴ CABG was performed according to the best clinical standards and left internal mammary artery grafting was strongly

recommended. In the PCI group, usage of intravascular ultrasound was recommended before and after the procedure but not mandated. Patients with ostial and mid-shaft lesions received single-stent techniques, whereas treatment of the left main coronary artery bifurcation could be performed with single-stent and two-stent techniques with mandatory kissing-balloon inflation. If a two-stent technique was performed, we recommended using the culotte technique. Patients received dual antiplatelet therapy for 12 months in both groups with a P2Y12 inhibitor and aspirin if the indication for revascularisation was acute coronary syndrome. PCI-treated patients with chronic coronary syndrome received 12 months of dual antiplatelet therapy. Adjudication of SYNTAX score was performed by an independent core laboratory (European Cardiovascular Research Center, Massy, France). The trial included a prespecified 10-year follow-up mandated as per protocol.

Outcomes

The prespecified primary outcome was the difference in long-term 10-year all-cause mortality. Vital status was assessed using a medical record audit or an active health insurance at the time of the audit at one participating site. Serious adverse events and adverse events (other than mortality) were not assessed at 5–10 years; therefore, the adverse event table is not reported here.

Statistical analysis

Power analysis of the primary endpoint has been published previously.⁴ The statistical analysis plan for the 10-year endpoint was not prespecified in the protocol; therefore, for consistency, the analyses were conducted in accordance with the original analysis plan used for the 5-year endpoint. Width of CIs of secondary endpoints have not been adjusted for multiplicity and should not be used to infer treatment effects. Continuous variables are reported as mean (SD) or median (IQR) dependent on distribution and calculated using the Student's *t* test or Mann–Whitney test. Binary and categorical variables are reported as counts (%) and calculated using the χ^2 test. Follow-up began at randomisation and patients were censored at the date of death, follow-up, emigration, or loss to follow-up (whichever occurred first). The difference between study groups in the endpoint of all-cause mortality was compared using survival analysis. Kaplan–Meier time-to-event curves were plotted, and the difference was calculated using the log-rank test. Cox proportional hazards were used to calculate the difference in hazard during the 10-year period and are reported as hazard ratios (HRs) with 95% CIs. The proportional hazards assumption was tested using log–log plots of observed versus predicted values. Cox regression was adjusted for clinically prespecified stratification variables of sex, distal left main coronary artery bifurcation lesions, and diabetes.

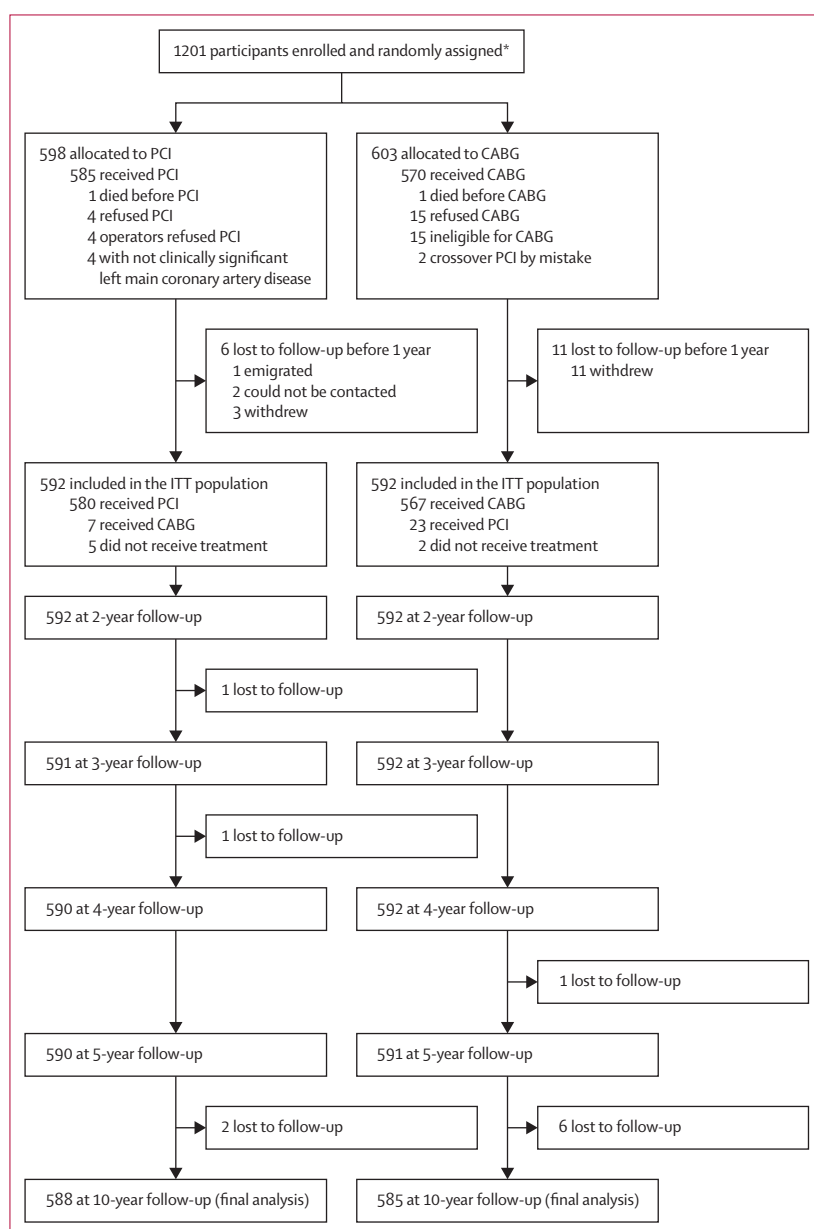


Figure 1: Trial profile

CABG=coronary artery bypass grafting. ITT=intention-to-treat. PCI=percutaneous coronary intervention.

*Adapted from Holm et al.³

A sensitivity landmark post-hoc subgroup analysis of 0–5 years and 5–10 years was stratified by sex (male vs female), age at randomisation (quartiles), indication (acute or chronic coronary syndrome), diabetes, distal left main coronary artery bifurcation, lesions treated, BMI, and SYNTAX score (≤ 22 , 23–32, and ≥ 33). The intention-to-treat (ITT) principle was used for all analyses. In a sensitivity analysis, we fitted a Bayesian–Weibull proportional hazards model using an informative prior for the treatment effect based on external data from the SYNTAX and PRECOMBAT trials.^{7,8} The prior for treatment HR was 1·10, corresponding to a normal prior on the log HR: normal (mean 0·09531 [SD 0·0834]). Weakly informative priors were used for the baseline hazard (shape parameter). Posterior estimates were obtained using Markov chain Monte Carlo with 12 500 iterations, including a burn-in of 2500 iterations. A two-sided p value of less than 0·05 was considered statistically significant. All analyses were performed using Stata (version 19).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

From Dec 9, 2008, to Jan 21, 2015, 1201 patients were randomly assigned to PCI (n=598) or CABG (n=603). 17 patients were lost to follow-up before 1 year (11 withdrew consent in the CABG group and three withdrew consent, one emigrated, and two could not be contacted in the PCI group; figure 1). 592 patients in each group were included in the ITT population. At 10 years, vital status was available in 1173 (98%) patients (1170 collected via medical record audit and three with an active health insurance at the time of the audit). The trial profile and CONSORT flow chart have previously been published.^{3,4}

Baseline characteristics have previously been reported.⁴ Mean age was 66·2 years (SD 9·9) in the PCI group and 66·2 years (9·4) in the CABG group. 256 (22%) of 1184 participants were female and 928 (78%) were male. The logistic European System for Cardiac Operative Risk Evaluation was 2 (IQR 2–4) in both groups, core laboratory SYNTAX score was 22·4 (SD 7·8) in the PCI group versus 22·3 (7·4) in the CABG group, and the presence of a distal left main coronary artery lesion was also well balanced between the groups (477 [81%] vs 482 [81%]; table). Surgical procedural characteristics are shown in the appendix (p 6). The median time to procedure in the CABG group was 6 days (IQR 2–10) in patients with acute coronary syndrome and 17 days (7–40) in those with chronic coronary syndrome, whereas in the PCI group, the median time to procedure was 1 day (0–3) versus 6 days (1–15; appendix p 11).

The Kaplan–Meier estimate of all-cause mortality at 10 years was 23% (136 of 592) in the PCI group and 25% (145 of 592) in the CABG group and was not statistically different between the groups (HR 0·93 [95% CI 0·74–1·18]; p=0·56). In a time-restricted landmark analysis, all-cause mortality occurred in 9% in both groups at 0–5 years (HR 1·06 [0·73–1·55]; p=0·76). At 5–10 years, the estimates were 15% in the PCI group and 17% in the CABG group (HR 0·86 [0·64–1·16]; p=0·32; figure 2). Subgroup analyses are presented in figure 3. An interaction with indication was observed in patients with acute coronary syndrome (HR 0·57 [0·32–0·99]) and chronic coronary syndrome (HR 1·04 [0·80–1·34]; $p_{\text{interaction}}=0·049$; figure 4). No interaction with SYNTAX score was observed, but HRs were leaning in opposite directions for low and intermediate SYNTAX score (figure 5). Baseline characteristics for the three SYNTAX groups and individual major adverse cardiovascular and

	PCI group (n=592)	CABG group (n=592)	p value
Age, years	66·2 (9·9)	66·2 (9·4)	0·91
Sex			
Female	116 (20%)	140 (24%)	0·09*
Male	476 (80%)	452 (76%)	..
BMI, kg/m ²	27·9 (4·5)	28·1 (4·4)	0·53
Type 1 or type 2 diabetes	86 (15%)	90 (15%)	0·94
Family history of ischaemic heart disease	321 (58%)	307 (56%)	0·45
Statin treatment	482 (82%)	464 (78%)	0·17
Hypertension	386 (65%)	489 (66%)	0·91
Active smoking	108 (19%)	127 (22%)	0·18
Previous PCI	116 (20%)	118 (20%)	0·90
Previous CABG	4 (1%)	2 (<1%)	0·41
Ejection fraction	60 (55–65)	60 (52–64)	0·27
New York Heart Association Functional Classification			
I	244 (53%)	195 (43%)	..
II	135 (29·6%)	150 (33%)	..
III	57 (13%)	77 (17%)	..
IV	23 (5%)	33 (7%)	0·012
European System for Cardiac Operative Risk Evaluation	2 (2–4)	2 (2–4)	0·18
SYNTAX score	22·4 (7·8)	22·3 (7·4)	0·74
Indication			
Stable angina pectoris	486 (82%)	491 (83%)	0·66
Acute coronary syndrome	106 (18%)	100 (17%)	0·66
Number of lesions requiring treatment	2 (1–3)	2 (2–3)	<0·0001
Distal left main coronary artery lesion	477 (81%)	482 (81%)	0·77

Data are mean (SD), n (%), median (IQR), or n (IQR). CABG=coronary artery bypass grafting. PCI=percutaneous coronary intervention. *p value shows the overall difference between female and male patients.

Table: Baseline characteristics

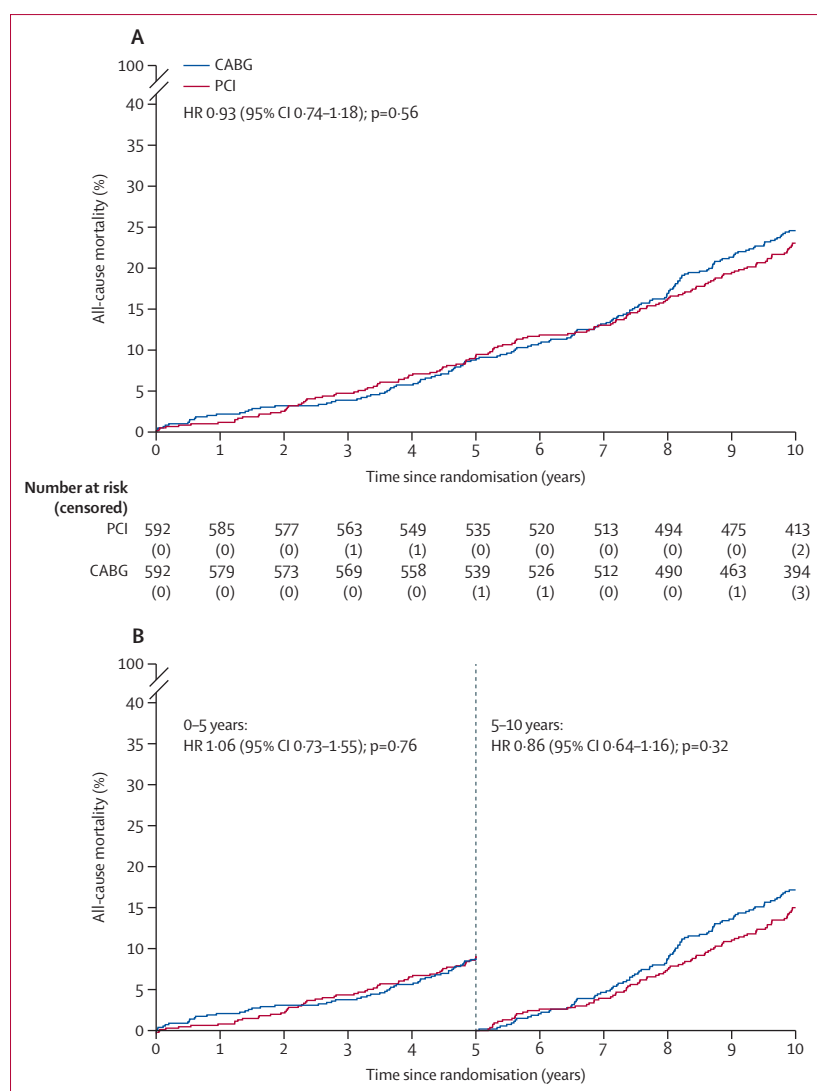


Figure 2: Primary endpoint and time-dependent landmark analysis

(A) Cumulative incidence of all-cause mortality. (B) Landmark analysis at 0–5 years and 5–10 years for all-cause mortality. CABG=coronary artery bypass grafting. HR=hazard ratio. PCI=percutaneous coronary intervention.

cerebrovascular event components at 5 years are shown in the appendix (pp 7–10). In terms of composite major adverse cardiovascular and cerebrovascular events at 5 years, there was no difference between study groups in those with acute coronary syndrome (HR 1.00 [0.59–1.71]); whereas a significant increase was observed in those with chronic coronary syndrome (HR 1.72 [1.32–2.25]). Baseline characteristics stratified by indication are shown in the appendix (p 11).

The Kaplan–Meier estimate of all-cause mortality at 10 years in patients who received first generation DES was 35% (16 of 46) and was 22% (116 of 524) among those who received second generation DES. Baseline differences between first generation and second generation DES and the primary endpoint at 5 years are

shown in the appendix (pp 13–14). Sensitivity analysis of all-cause mortality stratified by the inclusion period into quartiles did not show a heterogeneous treatment effect over time (appendix p 15). The Kaplan–Meier estimate of all-cause mortality was 20% (85 of 430) in patients who underwent intravascular ultrasound after PCI and 31% (45 of 144) in those who did not (appendix p 18). Baseline characteristics of patients with final intravascular ultrasound have previously been published (appendix p 16).¹¹ Use of intravascular ultrasound was lowest in the first and last calendar year quarters of the inclusion period (96 [62%] of 156 in quarter 1, 124 [86%] of 145 in quarter 2, 108 [75%] of 144 in quarter 3, and 102 [69%] of 147 in quarter 4). In a post-hoc sensitivity analysis with three exposed groups, the risk of all-cause mortality was HR 0.79 (95% CI 0.61–1.04) in patients who underwent intravascular ultrasound after PCI and HR 1.31 (0.94–1.83) in those who did not compared with the reference CABG group. Patients with complete arterial revascularisation in the CABG group had a lower all-cause mortality at 10 years (160 [28%] of 570) than those with incomplete arterial revascularisation (26 [16%] of 160 vs 92 [26%] of 361), which was consistent with findings at 5 years (appendix p 17).

The Bayesian–Weibull sensitivity analysis, incorporating prior information corresponding to an HR of 1.10, yielded a posterior median HR of 0.95 (95% credible interval 0.74–1.18), which was consistent with the main analysis, indicating no substantial deviation from the primary finding and suggesting robustness to modelling assumptions and prior beliefs (the prior for treatment HR was 1.10).

Discussion

In this long-term follow-up analysis of the NOBLE trial, we found no difference in all-cause mortality 10 years after PCI or CABG in patients with unprotected left main coronary artery disease and no additional complex lesions. In a time-dependent landmark analysis there was no difference between the groups at 0–5 years or 5–10 years. A prespecified subgroup analysis showed a differential treatment effect by indication.

These findings are consistent with a sensitivity analysis from a meta-analysis of long-term outcomes, including 10-year follow-up of 1305 patients who received first generation DES in the SYNTAX Extended Survival and PRECOMBAT trials.⁶ In the only other second generation DES trial, EXCEL showed a significantly increased mortality rate 5 years after random assignment of patients who underwent PCI (OR 1.38 [95% CI 1.03–1.85]).⁵ This finding contributed to a strong association with higher all-cause mortality in the meta-analysis of randomised controlled trials on the left main coronary artery treatment strategy.⁶ A late mortality catch-up among patients who underwent PCI in NOBLE might have been expected; however, this hypothesis was not supported by our 5–10-year landmark

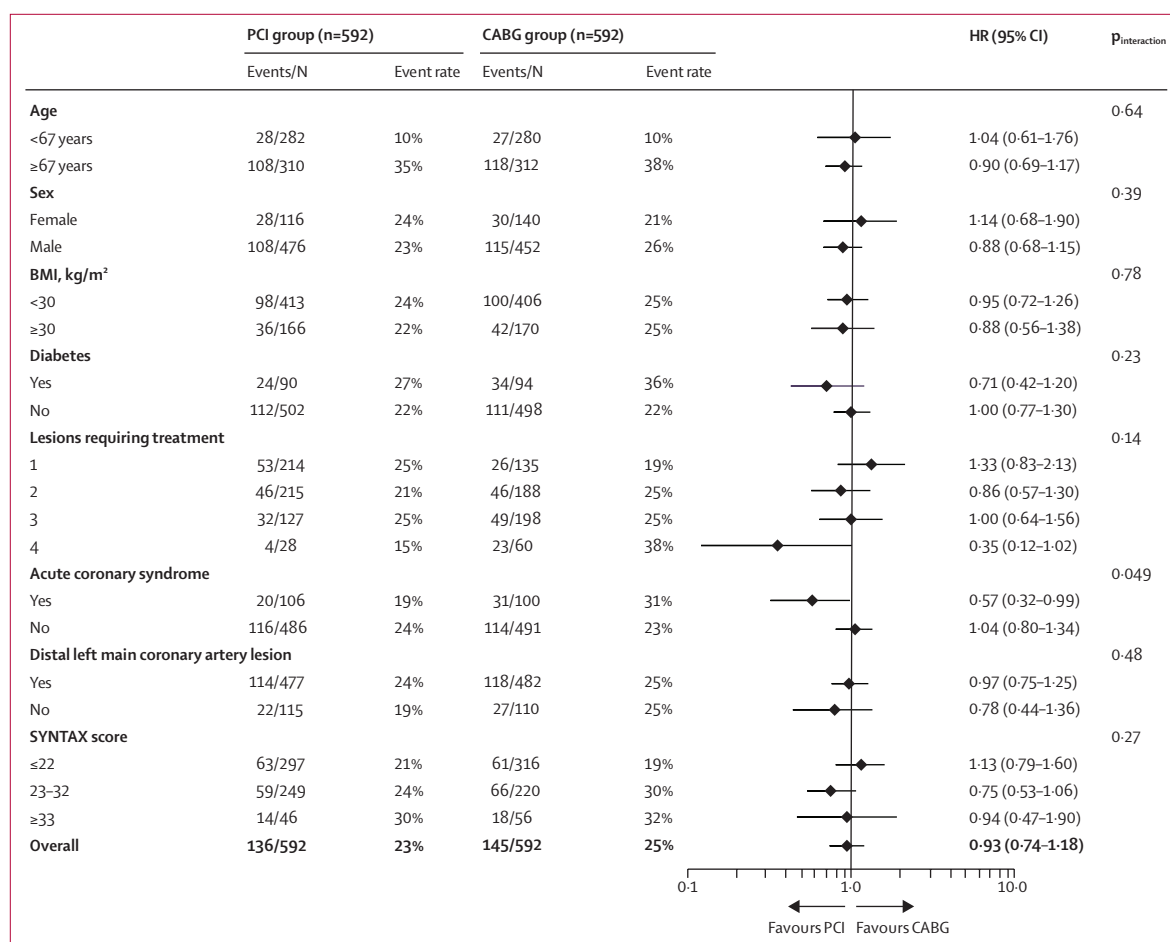


Figure 3: Forest plot of prespecified subgroups

CABG=coronary artery bypass grafting. HR=hazard ratio. PCI=percutaneous coronary intervention.

analysis. Our findings, in combination with previously published results, support the guideline recommendations that PCI can be a safe alternative to CABG in patients with unprotected left main coronary artery disease;¹ however, CABG remains superior for reducing spontaneous myocardial infarction and repeat revascularisation, but carries a higher risk of early stroke.⁶ Importantly, this increased risk does not translate into a mortality difference, which has been inconsistently reported.¹²⁻¹⁵ Caution is needed regarding interpretation of the first generation DES results because of the higher all-cause mortality rate seen in these patients compared with those who received second generation DES (16 [35%] of 46 vs 116 [22%] of 524). This difference might be due to chance, as excess mortality with PCI versus CABG was not observed in PRECOMBAT or SYNTAX Extended Survival. The excess mortality with first generation DES could also be due to a higher number of fatal myocardial infarctions as the 5-year myocardial infarction rates were similar, but mortality significantly higher and first generation

DES are more prone to stent thrombosis.¹⁶ Furthermore, we find that the higher mortality rates early in the study did not influence differences between the study groups, which is highlighted by a sensitivity analysis accounting for the inclusion period. The results from NOBLE reassure us that second generation stents are safe to use in patients with left main coronary artery disease. This finding emphasises that shared decision making with the patient and heart team is crucial, a principle strongly recommended in current guidelines.¹ Our results will inform and enlighten this practice for physicians and patients.

A tool recommended to aid the heart team in clinical decision making is the SYNTAX score;¹ however, the previously reported numerical suggestion of a benefit of CABG in patients with left main coronary artery disease and high SYNTAX score was not observed in NOBLE, although this subgroup was small. We find that treatment effect in low, intermediate, or high SYNTAX score is similar, which is consistent with the 5-year pooled all-cause mortality⁶ and 10-year findings in PRECOMBAT

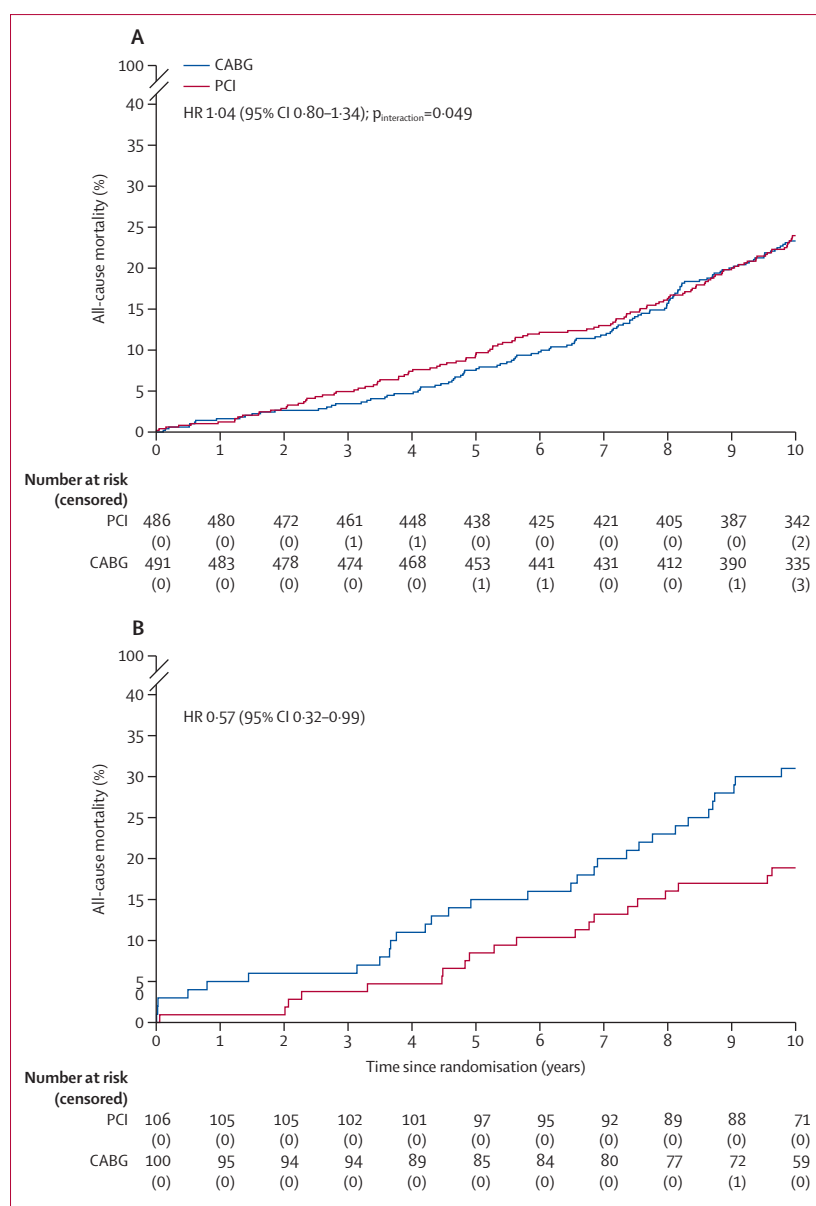


Figure 4: All-cause mortality stratified by treatment indication

Cumulative incidence of all-cause mortality in patients with chronic coronary syndrome (A) and acute coronary syndrome (B). CABG=coronary artery bypass grafting. HR=hazard ratio. PCI=percutaneous coronary intervention.

and SYNTAX Extended Survival.^{6,17,18} Additionally, a high SYNTAX score was not predictive of major adverse cardiac events in EXCEL or NOBLE—in fact, low SYNTAX score favoured CABG in NOBLE.^{3,5} The guideline recommendation^{1,2} might require a revision and we believe that in patients with left main coronary artery disease (without additional complex lesions), the SYNTAX score should be applied cautiously for clinical decision making within the heart team.

The only significant second generation stents treatment effect was found in those with acute coronary syndrome, which favoured PCI; whereas, a 2023

meta-analysis of patients with acute or chronic coronary syndrome and left main coronary artery disease found no heterogeneous treatment effect on all-cause mortality, spontaneous myocardial infarction, or repeat revascularisation at 5 years.¹⁶ In NOBLE, we previously reported a heterogeneous treatment effect of major adverse cardiac and cerebrovascular events at 5 years and in the present study, the 10-year all-cause mortality shows a consistent signal. A possible explanation for the different 10-year results might be an effect of earlier revascularisation, which is associated with improved outcomes.^{17,18} Importantly, no patients with ST elevation myocardial infarction were included in our study, and the time between random assignment and revascularisation in patients with acute coronary syndrome was 6 days (IQR 2–10) in the CABG group versus 1 day (0–3) in the PCI group—indicating a delay in treatment in the CABG group. In our analysis, the interaction between indication and treatment was mostly driven by a higher event rate in the CABG group than that seen in other trials (31% in NOBLE vs 22% in SYNTAX Extended Survival and PRECOMBAT).¹⁶ Nonetheless, guideline recommendations for revascularisation of patients with acute coronary syndrome are vague and based on underpowered trials (NOBLE, EXCEL, SYNTAX, and PRECOMBAT).^{6,19} Our results indicate that PCI could be considered over CABG for patients with acute coronary syndrome and significant left main coronary artery disease, particularly in Europe where the strategy is often chosen for older clinically unstable patients at high surgical risk.

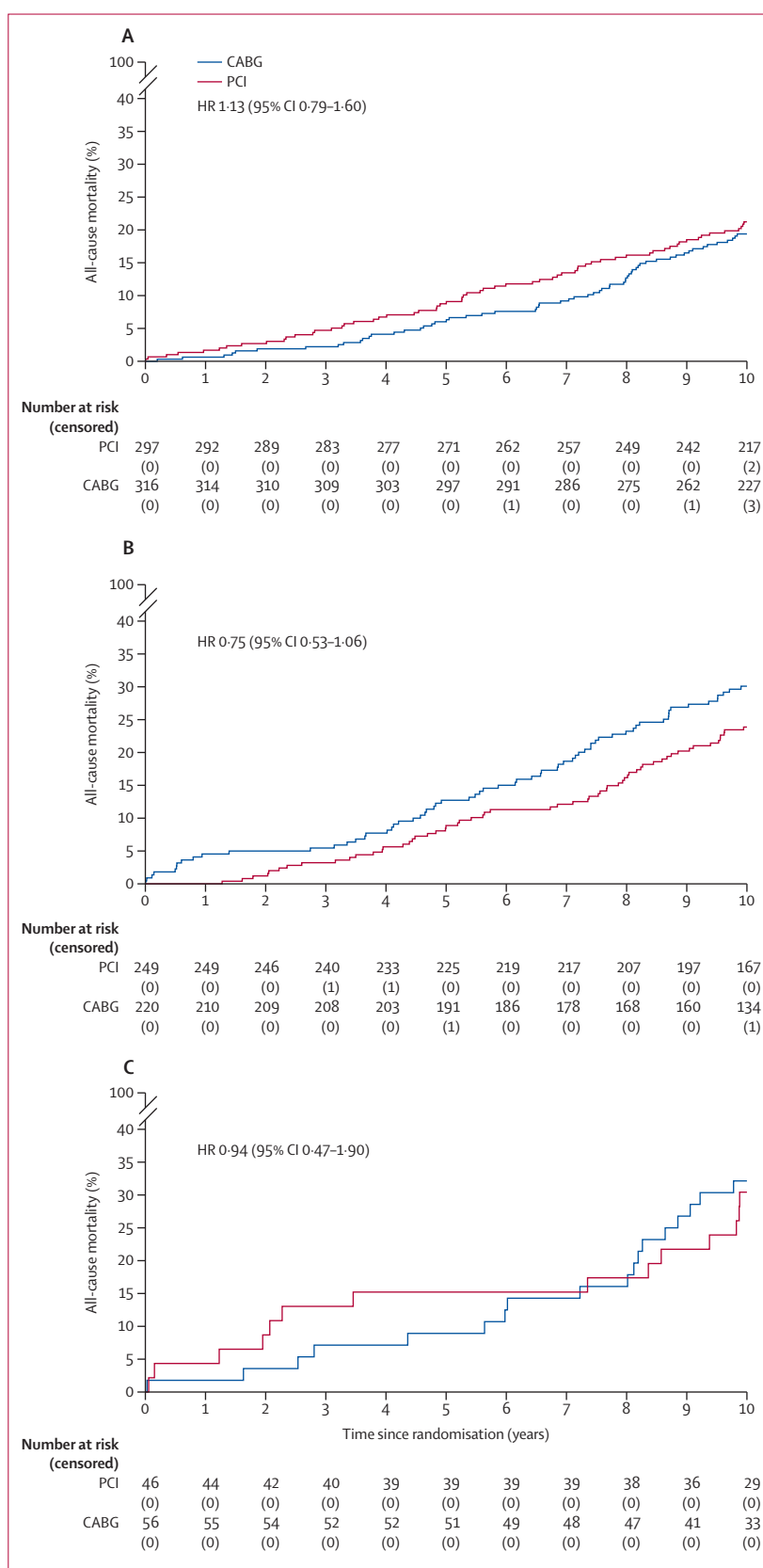
The current analysis is limited by an absence of secondary endpoints, such as myocardial infarction, stroke, repeat revascularisation, and cause of death. Additionally, the enrolment period (2008–15) might introduce temporal confounding. Modern PCI using intravascular imaging and newer generation DES was not used in our study. In complex lesions, intravascular imaging has been associated with improved short-term major adverse cardiac events and reduced all-cause mortality.^{20,21} From previous sub-studies of NOBLE, intravascular ultrasound after the procedure and appropriate stent expansion were associated with better outcome in terms of major adverse cardiac and cerebrovascular events.¹¹ These observations are supported by a sensitivity analysis in our study as final intravascular ultrasound control was associated with a numerically lower event rate in patients who underwent PCI (all-cause mortality 20% [85 of 430] vs 31% [45 of 144] with no final intravascular ultrasound) and no final intravascular ultrasound was associated with more deaths than CABG. This analysis is limited by a difference in age between patients who underwent intravascular ultrasound and no intravascular ultrasound, and no detailed prespecified intravascular ultrasound protocol was defined.¹¹ The study is also limited by advances in guideline-directed medical

therapy for secondary prevention of ischaemic events, which have substantially evolved and might influence all-cause mortality in both groups as the usage of statins at random assignment was around 80% in both groups.^{1,22} Adherence to medical therapy was not assessed. In addition, the power in the analysis of heterogeneous treatment effects across subgroups is limited by low numbers, which is most pronounced in patients with high SYNTAX score. Importantly, these results do not apply to patients with left main coronary artery disease and additional complex lesions as this population was in the exclusion criteria. Finally, all participants included in our study were White and the external validity beyond this population might be limited as NOBLE was conducted in northern Europe; however, the conclusions are supported by findings from the PRECOMBAT trial performed in South Korea.⁹

Statistical testing of all-cause mortality was not powered before study initiation and hypothesis testing should be interpreted with caution. Although the point estimate is close to 1 and 95% CIs are narrow, a type 2 error cannot be excluded. Nevertheless, the Bayesian analyses, including those incorporating informative priors, support the conclusion of no meaningful difference between the PCI and CABG groups. No formal prespecified analysis plan addressing testing for multiplicity was performed. Accordingly, findings in the acute coronary syndrome subgroup should be interpreted with caution, as the risk of a type 1 error cannot be excluded.

Despite limitations, this long-term follow-up study might affect future guidelines for patients with acute and chronic coronary syndrome by supporting more nuanced patient-centred decisions. Notably, these guidelines need to emphasise that CABG remains superior for major adverse cardiac events, although modern PCI warrants further evaluation. As seen in the FAME 3 trial, a 5-year secondary analysis found no difference between fractional flow reserve-guided PCI and CABG in patients with three vessel disease on a composite of all-cause death, myocardial infarction, and stroke, contrasting the previous SYNTAX trial findings.^{23,24} Explorative analyses in the RENOVATE COMPLEX trial of PCI and NOBLE intravascular ultrasound study suggest that intravascular imaging-guided PCI might reach equivalent outcomes to CABG in patients with left main coronary artery disease.^{3,11,25} The hypotheses generated by these studies require validation in adequately powered trials with long-term follow-up.

Figure 5: All-cause mortality stratified by SYNTAX score
Cumulative incidence of all-cause mortality in patients with a low SYNTAX score of 1–22 (A), intermediate SYNTAX score of 23–32 (B), and a high SYNTAX score of at least 33 (C). CABG=coronary artery bypass grafting. HR=hazard ratio. PCI=percutaneous coronary intervention.



No difference in all-cause mortality was found at 10 years between CABG and PCI for patients with unprotected left main stenosis and no additional complex lesions, indicating that PCI is equally as safe as CABG in patients eligible for both treatments. These results will aid heart teams in developing an individualised patient-centred strategy and inform shared decision making and future guideline recommendations.

Contributors

LT designed the trial and wrote the protocol. Contracts and funding were overseen by JFL, LT, EHC, and NRH. Statistical analysis was done by LJHM and ENH. LJHM and ENH have assessed and verified all the data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication. The writing committee was led by ENH and consisted of NRH, DH-S, MML, MSS, AE, IBAM, ISM, MN, JFL, TT, KO, PS, SJW, PCE, ANJG, VA, UT, LT, and EHC. The writing committee interpreted results and wrote the report. All authors contributed to implementation of the study and data acquisition, and approved the manuscript for publication.

Declaration of interests

ENH received institutional grants from Abbott, Phillips, Orbus Neich, and Asahi Intecc. EHC received institutional grants from Abbott, Phillips, Orbus Neich, Biosensors, and Asahi Intecc; and personal fees from Abbott for teaching. NRH is an employee of Novo Nordisk; and received institutional grants from Abbott, Biosensors, and Boston Scientific. OF received speaker honoraria from Sanofi Pasteur and Pfizer; and consultancy fees from GSK. LT is an employee of Edwards Lifescience. All other authors declare no competing interests.

Data sharing

Individual patient data will not be made publicly available due to the applicable law. De-identified participant data will be made available upon reasonable request to the corresponding author after approval of a proposal, a signed data agreement, and approval by the principal investigator.

Acknowledgments

This study was funded by Biosensors and Aarhus University Hospital and participating sites. We thank Helle Bargsteen for being a study secretary and administrative cornerstone since study initiation and Jakob Hjorth for establishing and maintaining the study database.

References

- Vrints C, Andreotti F, Koskinas KC, et al. 2024 ESC guidelines for the management of chronic coronary syndromes. *Eur Heart J* 2024; **45**: 3415–537.
- Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology Joint Committee on clinical practice guidelines. *Circulation* 2023; **148**: e9–119.
- Holm NR, Mäkilä T, Lindsay MM, et al. Percutaneous coronary angioplasty versus coronary artery bypass grafting in the treatment of unprotected left main stenosis: updated 5-year outcomes from the randomised, non-inferiority NOBLE trial. *Lancet* 2020; **395**: 191–99.
- Mäkilä T, Holm NR, Lindsay M, et al. Percutaneous coronary angioplasty versus coronary artery bypass grafting in treatment of unprotected left main stenosis (NOBLE): a prospective, randomised, open-label, non-inferiority trial. *Lancet* 2016; **388**: 2743–52.
- Stone GW, Kappetein AP, Sabik JF, et al. Five-year outcomes after PCI or CABG for left main coronary disease. *N Engl J Med* 2019; **381**: 1820–30.
- Sabatine MS, Bergmark BA, Murphy SA, et al. Percutaneous coronary intervention with drug-eluting stents versus coronary artery bypass grafting in left main coronary artery disease: an individual patient data meta-analysis. *Lancet* 2021; **398**: 2247–57.
- Ahn JM, Roh JH, Kim YH, et al. Randomized trial of stents versus bypass surgery for left main coronary artery disease: 5-year outcomes of the PRECOMBAT Study. *J Am Coll Cardiol* 2015; **65**: 2198–206.
- Mohr FW, Morice M-C, Kappetein AP, et al. Coronary artery bypass graft surgery versus percutaneous coronary intervention in patients with three-vessel disease and left main coronary disease: 5-year follow-up of the randomised, clinical SYNTAX trial. *Lancet* 2013; **381**: 629–38.
- Park D-W, Ahn J-M, Park H, et al. Ten-year outcomes after drug-eluting stents versus coronary artery bypass grafting for left main coronary disease: extended follow-up of the PRECOMBAT Trial. *Circulation* 2020; **141**: 1437–46.
- Thuijs DJFM, Kappetein AP, Serruys PW, et al. Percutaneous coronary intervention versus coronary artery bypass grafting in patients with three-vessel or left main coronary artery disease: 10-year follow-up of the multicentre randomised controlled SYNTAX trial. *Lancet* 2019; **394**: 1325–34.
- Ladwiniec A, Walsh SJ, Holm NR, et al. Intravascular ultrasound to guide left main stem intervention: a NOBLE trial substudy. *EuroIntervention* 2020; **16**: 201–09.
- O'Fee K, Deych E, Ciani O, Brown DL. Assessment of nonfatal myocardial infarction as a surrogate for all-cause and cardiovascular mortality in treatment or prevention of coronary artery disease: a meta-analysis of randomized clinical trials. *JAMA Intern Med* 2021; **181**: 1575–87.
- Palmerini T, Della Riva D, Biondi-Zoccai G, et al. Mortality following nonemergent, uncomplicated target lesion revascularization after percutaneous coronary intervention: an individual patient data pooled analysis of 21 randomized trials and 32,524 patients. *JACC Cardiovasc Interv* 2018; **11**: 892–902.
- Kim TO, Kang D-Y, Ahn J-M, et al. Impact of target lesion revascularization on long-term mortality after percutaneous coronary intervention for left main disease. *JACC Cardiovasc Interv* 2024; **17**: 32–42.
- Giustino G, Serruys PW, Sabik JF 3rd, et al. Mortality after repeat revascularization following PCI or CABG for left main disease: the EXCEL trial. *JACC Cardiovasc Interv* 2020; **13**: 375–87.
- Gaba P, Christiansen EH, Nielsen PH, et al. Percutaneous coronary intervention vs coronary artery bypass graft surgery for left main disease in patients with and without acute coronary syndromes: a pooled analysis of 4 randomized clinical trials. *JAMA Cardiol* 2023; **8**: 631–39.
- Diletti R, den Dekker WK, Bennett J, et al. Immediate versus staged complete revascularisation in patients presenting with acute coronary syndrome and multivessel coronary disease (BIOVASC): a prospective, open-label, non-inferiority, randomised trial. *Lancet* 2023; **401**: 1172–82.
- Marquard JM, Beske RP, Kelbæk H, et al. 10-year outcome of complete or infarct artery-only revascularization in STEMI with multivessel disease: the DANAMI-3-PRIMULTI study. *J Am Coll Cardiol* 2025; **86**: 119–29.
- Byrne RA, Rossello X, Coughlan JJ, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. *Eur Heart J* 2023; **44**: 3720–826.
- Stone GW, Christiansen EH, Ali ZA, et al. Intravascular imaging-guided coronary drug-eluting stent implantation: an updated network meta-analysis. *Lancet* 2024; **403**: 824–37.
- Holm NR, Andreasen LD, Neghabat O, et al. OCT or angiography guidance for PCI in complex bifurcation lesions. *N Engl J Med* 2023; **389**: 1477–87.
- Mach F, Koskinas KC, Roeters van Lennep JE, et al. 2025 focused update of the 2019 ESC/EAS guidelines for the management of dyslipidaemias. *Eur Heart J* 2025; **46**: 4359–78.
- Fearon WF, Zimmermann FM, Ding VY, et al. Outcomes after fractional flow reserve-guided percutaneous coronary intervention versus coronary artery bypass grafting (FAME 3): 5-year follow-up of a multicentre, open-label, randomised trial. *Lancet* 2025; **405**: 1481–90.
- Serruys PW, Morice M-C, Kappetein AP, et al. Percutaneous coronary intervention versus coronary-artery bypass grafting for severe coronary artery disease. *N Engl J Med* 2009; **360**: 961–72.
- Lee SY, Cho YH, Sung K, et al. Intravascular imaging-guided PCI vs coronary artery bypass grafting for left main or 3-vessel disease. *JACC Cardiovasc Interv* 2025; **18**: 2077–88.